

Arun Rimal

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SUMMARY

Software Engineer & AI/ML Engineer with an M.S. in Data Science (GPA 3.8) and significant years of professional experience building production-grade full-stack applications, backend services, and AI-powered systems. Proficient in Python, TypeScript/Node.js, React/Next.js, FastAPI, GraphQL, and Docker with experience in cross-functional teams. Hands-on experience in LLM application development, transformer fine-tuning, RAG architectures, and agentic AI workflows using LangChain, LlamaIndex, LangGraph, and major LLM APIs (OpenAI, Gemini, Claude).

PROFESSIONAL EXPERIENCE

Graduate AI/ML Researcher

Jan 2025 – May 2026

NIAR, Wichita State University, Wichita, KS

- Developed and implemented secure coding practices in a multi-user PDF Q&A RAG Application using FastAPI, Next.js, and LlamaIndex, leveraging Google Gemini for reasoning and Pinecone for vector storage to enable real-time, context-aware document analysis with isolated user sessions.

Software Engineer, Data

Feb 2024 – Jul 2024

Infinite Computer Solutions, Kathmandu, Nepal

- Developed and maintained ETL pipelines and stored procedures using MSSQL and SSIS to process US healthcare data across multiple client environments.
- Refactored legacy procedures, improving pipeline reliability and reducing manual intervention in data transformation workflows.

Data Engineer / Backend Developer

Sep 2021 – Feb 2024

Devfinitly Nepal, Kathmandu, Nepal

- Built a Python/Flask ETL microservice integrating 6+ data sources (REST APIs, SQL Server, Azure Blob, SFTP, Google Analytics 4) into a multi-tenant BI data warehouse with Kafka-based async processing and Dockerized deployment.
- Developed an Identity & Account Management Module with JWT-based authentication, role-based authorization (Super-Admin, Tenant-Admin, App User), token refresh flows, and multi-tenant user isolation using Node.js, TypeScript, and GraphQL.
- Designed end-to-end Azure Data Factory pipelines (Transact → Staging → Production) improving data freshness for downstream BI reporting.
- Developed and optimized stored procedures for data transformation and workflow automation, reducing ad-hoc query load on production systems.

Data Analyst — Network Operations

Aug 2015 – Aug 2021

Huawei Technologies Nepal, Lalitpur, Nepal

- Processed 2G/3G/4G RF drive test logs using TEMS Investigation, exporting to CSV/DBF for KPI analysis (MOS, Throughput, RSCP, RSSI) in Excel to support network optimization.
- Visualized geospatial drive test coverage in MapInfo, overlaying site databases and drive routes to prepare benchmark reports comparing Ncell and Nepal Telecom networks.
- Produced weekly power and MW performance reports using Excel, delivering data-driven insights to support Ncell network operations and optimization decisions.

PROJECTS

- **MLOps Pipeline** Designed and implemented an end-to-end MLOps pipeline for employee attrition prediction using Python, scikit-learn, and Feast Feature Store for point-in-time feature engineering. Performed model training, evaluation, and hyperparameter tuning (GridSearchCV) across multiple classification algorithms. Leveraged MLflow for experiment tracking, model versioning, and artifact management, and implemented CI/CD-style model deployment using a model registry with Staging/Production workflows. Integrated Slack webhooks for automated pipeline monitoring and notifications.
- **Financial Sentiment Analysis (ProsusAI/FinBERT)** Fine-tuned a transformer-based FinBERT model for financial sentiment classification, achieving 87% accuracy and outperforming baseline models including original FinBERT, DistilBERT, LSTM, and Naive Bayes.
- **Credit Risk Modeling**
Built loan default prediction models using XGBoost, Logistic Regression, and Random Forest; XGBoost achieved up to 98% accuracy across cross-validated evaluation sets to support credit risk decisioning.

- **Kansas BWA NLP Pipeline**
Built a production-grade NLP pipeline with Apache Airflow, custom spaCy NER, Gemini LLM classification, and fuzzy string matching to extract and classify public health text data from KDHE, with downstream geospatial analysis.
- **AWS ETL Pipeline for CSV and API Data**
Developed an automated AWS ETL pipeline to extract and validate data from CSV files and REST APIs, event-triggered via AWS SQS.
- **AI-Powered Tableau Dashboard Generator** (*In Progress*)
LLM reasoning pipeline (dataset analysis → KPI recommendation → Tableau dashboard generation) combining agentic LLM workflows with BI tooling.

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Node.js
Frontend	React, Next.js, HTML/CSS
Backend & APIs	FastAPI, Flask, REST APIs, GraphQL, SQLAlchemy, Prisma
Data & ETL	Pandas, NumPy, PySpark, Apache Airflow, Azure Data Factory, AWS, Snowflake, SSIS
ML & Deep Learning	Scikit-learn, TensorFlow, Keras, PyTorch, XGBoost, CNNs, RNNs, LSTMs, Transformers, MLflow
NLP	spaCy, HuggingFace Transformers, FinBERT, NER, Sentiment Analysis
LLM & Agentic AI	LangChain, LangGraph, LlamaIndex, OpenAI API, Claude API, Gemini API, RAG, LLM Fine-tuning
Databases	PostgreSQL, MSSQL, MySQL, MongoDB, Pinecone, ChromaDB
Visualization	Tableau, PowerBI, Matplotlib, Seaborn, GeoPandas
Cloud & DevOps	AWS (S3, SQS, Lambda), GCP(VM, cloud Storage), Azure, Docker, version control - Git, GitHub, Jenkins(CI/CD), Terraform(IaC)
Auth & Security	JWT, OAuth2
Other	Agile, generative AI, neural networks, LLMs

EDUCATION

- **Master of Science, Data Science** **Aug 2024 – 2026**
Wichita State University, Wichita, Kansas — *GPA: 3.8/4.0*
- **Master of Business Administration** **Aug 2019 – 2023**
DAV Business School, Lalitpur, Nepal
- **Bachelor of Electronics and Communication Engineering** **Aug 2010 – 2014**
Kantipur Engineering College, Lalitpur, Nepal